

COURSE SYLLABUS

AWS Administration Essentials

COURSE NUMBER

AWS-101

DURATION

1 day · 4 modules

AUDIENCE LEVEL

Beginner

FORMAT

Instructor-Led / Self-Paced

Course syllabus

COURSE OVERVIEW

This one-day course teaches students the basics of creating and administering resources in the AWS Cloud. Students will learn to navigate and configure AWS environments using the AWS Management Console, Cloud Shell, and command-line tools.

DELIVERY FORMAT OPTIONS

Standard Instructor-Led Training (ILT). The instructor teaches each module, leads explanations and discussions, and guides students through the standard module labs using the published module timing.

Bootcamp / Lab-Centered Delivery. When Bootcamp Mode capstone labs are included, they provide an alternative delivery configuration, not additional required ILT lecture time. In this configuration, students primarily work through module labs and capstone lab scenarios while the trainer acts as a facilitator, answering questions, clarifying concepts, and helping students work through lab challenges.

Training providers may choose either the normal ILT model or the Bootcamp/lab-centered model based on audience readiness, delivery goals, and available class time.

AT COURSE COMPLETION

After completing this course, you will be able to:

Navigate and interact with AWS resources using the Console, AWS CLI, and Cloud Shell commands.

Build a secure Virtual Private Cloud network utilizing segmented subnets, gateways, and security groups.

Deploy, customize, and attach storage resources using EC2 virtual instances and secure S3 storage models.

Establish active systems oversight utilizing CloudWatch metric triggers, CloudTrail auditing, and billing alarms.

PREREQUISITES

- Basic understanding of general networking concepts, such as IP addressing and subnets
 - Familiarity with basic command-line navigation and operating system concepts
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COURSE OUTLINE

MODULE 1 Navigating the AWS Cloud Environment

105 min

Learn to access and operate the primary management interfaces of AWS. This module guides you through the Console, Cloud Shell, AWS CLI configurations, and primary identity configurations.

LESSONS

- Lesson 1:** Exploring the AWS Management Console Layout
- Lesson 2:** Introduction to AWS Cloud Shell Capabilities
- Lesson 3:** Configuring AWS CLI Access Keys and Profiles
- Lesson 4:** IAM Fundamentals for System Administrators
- Lesson 5:** Troubleshooting AWS CLI Connection Errors

LAB EXERCISES

- Exploring AWS Core Navigation and IAM Administration

AFTER COMPLETING THIS MODULE, YOU WILL BE ABLE TO

- Navigate the AWS Management Console to locate core administrative services
- Configure AWS CLI local credentials and region default values
- Execute basic read-only queries within AWS Cloud Shell
- Explain key-pair authentication and identity configurations

MODULE 2 Virtual Private Cloud (VPC) Networking Essentials

105 min

Discover how isolated virtual networks function on AWS. Gain skills in designing subnets, managing route pathways, and safeguarding traffic flow.

LESSONS

Lesson 1: VPC CIDR Architecture and Subnet Segmentation

Lesson 2: Deploying Internet Gateways and Route Tables

Lesson 3: Implementing Stateless Network ACL Rules

Lesson 4: Creating Stateful Security Group Policies

Lesson 5: Isolating Private Networks and VPC Endpoints

LAB EXERCISES

Building a Custom Two-Tier VPC Architecture

AFTER COMPLETING THIS MODULE, YOU WILL BE ABLE TO

Calculate VPC IPv4 CIDR allocations and subnet scopes

Construct public and private routing tables containing active gateway entries

Configure stateful security group rules protecting virtual boundaries

Distinguish between NACL stateless rules and Security Group stateful configurations

MODULE 3 Compute and Storage Provisioning

105 min

Understand core server instances and simple object storage configurations. Learn to launch EC2 instances and connect them to S3 resources securely.

LESSONS

Lesson 1: Selecting and Customizing Amazon EC2 Instances

Lesson 2: Configuring Amazon EBS Storage Volumes

Lesson 3: Structuring Amazon S3 Bucket Properties

Lesson 4: Drafting Secure S3 Bucket Policies

Lesson 5: Bootstrapping Compute with EC2 User Data

LAB EXERCISES

Compute and Storage Provisioning in AWS

AFTER COMPLETING THIS MODULE, YOU WILL BE ABLE TO

Provision a configured Linux EC2 instance from an Amazon Machine Image (AMI)

Attach elastic EBS volumes to existing compute targets

Create S3 buckets and apply strict access bucket policies

Utilize instance user data scripts to automate software setup steps

MODULE 4 Monitoring and Resource Management

105 min

Examine operational visibility, governance tracking, and cost baseline rules. Learn how to configure active notifications and explore cost alerting limits.

LESSONS

Lesson 1: Tracking Compute and Network CloudWatch Metrics

Lesson 2: Designing CloudWatch Alarms and SNS Topics

Lesson 3: Auditing Administrative Actions with CloudTrail

Lesson 4: Organizing Systems via Strategic Tagging Schemas

Lesson 5: Constructing Financial Alerts with AWS Budgets

LAB EXERCISES

Configuring CloudWatch Monitoring, Alarms, and CloudTrail Auditing

AFTER COMPLETING THIS MODULE, YOU WILL BE ABLE TO

Construct AWS CloudWatch dashboards to track vital service metrics

Configure CloudWatch metrics thresholds to trigger SNS notifications

Audit administrative API calls using CloudTrail log engines

Apply dynamic tagging schemas for granular billing structures

BOOTCAMP MODE — CAPSTONE LABS

Capstone labs are included for Bootcamp/lab-centered delivery. They are not additional required lecture time for Standard Instructor-Led Training. Use them when the client chooses a facilitated lab delivery model where the trainer coaches students through integrated practice scenarios.

One integrative Capstone Lab is generated per course day. Each capstone presents a new scenario requiring students to independently apply the combined skills from that day's modules — no procedure, no hints. Students determine their own approach. Capstone Labs are the afternoon activity in Bootcamp delivery and replace the lecture portion of standard Instructor-Led Training.

DAY 1 CAPSTONE Modules 1, 2, 3, 4

SKILLS TO BE INTEGRATED

Initialize Cloud Shell and execute administrative CLI queries to extract resource details

Deploy a network layout complete with private and public segments, modifying routing tables to restrict flow

Launch an EC2 computer instance and associate it with customized S3 data repositories

Construct custom CloudWatch alerts linked to immediate notification workflows based on specific thresholds

Capstone content is embedded in the student manuals and instructor guides — one section per day, after the final module for that day. A standalone Capstone Slides (.pptx) file is also included in the Exclusive download package.